Assignment 3 — Normalization

Project: College Management System | Database: college_management_system | SQL dialect: MySQL 8.0

Based on the seven-table college design supplied for this project. Example records and expected outputs are illustrative, not screenshots from MySQL Workbench.

Design analysis

The seven entities separate departmental details, students, courses, faculty, registrations, examinations and grades. Department_ID is stored as a foreign key in Student, Faculty and Course, avoiding repetition of the department name and location in every record.

Functional dependencies

Department: Department_ID -> Department_Name, Location, HOD

Student: Student_ID -> Name, Email, Phone, Department_ID, Admission_Year; Email -> Student_ID (if unique and non-null)

Course: Course_ID -> Course_Name, Credits, Department_ID

Faculty: Faculty_ID -> Name, Email, Phone, Department_ID

Enrollment: Enrollment_ID -> Student_ID, Course_ID, Semester, Enrollment_Date; (Student_ID,Course_ID,Semester) -> Enrollment_ID

Examination: Exam_ID -> Course_ID, Exam_Date, Exam_Type, Total_Marks

Result: Result_ID -> Exam_ID, Student_ID, Marks, Grade; (Exam_ID,Student_ID) -> Result_ID

Potential anomalies

Insertion anomaly: in a single combined Student-Course-Department table, a new course cannot be recorded before a student enrolls. The separated Course table avoids this.

Update anomaly: storing Department_Name in every student row means a department rename must update many rows; missing one leaves conflicting names.

Deletion anomaly: deleting the last enrollment in a denormalized combined table might also erase the only information about the course.

Redundancy and improvements

Keep Department, Course, Student, Faculty, Enrollment, Examination and Result as separate relations with primary and foreign keys. Enforce UNIQUE(Student_ID,Course_ID,Semester) and UNIQUE(Exam_ID,Student_ID) to prevent duplicate registrations and scores.

The original HOD field stores a name rather than a Faculty_ID reference; this can become inconsistent with Faculty records. A future enhancement can use nullable HOD_Faculty_ID with a foreign key and an additional rule to ensure the faculty member belongs to the department.

Result.Grade can be derived from Marks and Total_Marks, so storing both can introduce an update anomaly. Keep Grade as a recorded official grade only if institutional grading rules require it, otherwise calculate it in a

view or application logic.

Results currently link a student and exam but do not enforce that the student is enrolled in the exam course. This cross-table business rule needs a trigger or controlled application workflow.

Normalization conclusion

With the documented key dependencies, the seven-table design avoids major partial and transitive dependencies and is broadly consistent with 3NF under the stated assumptions. Additional business rules and grade handling should be finalized before production use.